

Part I

SUMMARY of Definite Integrals

Summary

February 21, 2026

Definition

If function f is continuous on interval $[a, b]$ then

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

Note 1: When $f(x) \geq 0$, on the interval $[a, b]$, the definite integral, $\int_a^b f(x) dx$, gives the **area** of the region between the graph of f and the interval $[a, b]$ on the x -axis.

Note 2: When $f(x) < 0$, on some interval in $[a, b]$, then the definite integral, $\int_a^b f(x) dx$, gives the **net area** of the region between the graph of f and the interval $[a, b]$ on the x -axis.

Properties of Definite Integrals

$$(a) \int_a^a f(x) dx = 0$$

$$(b) \int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$(c) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$(d) \int_a^b k \cdot f(x) dx = k \cdot \int_a^b f(x) dx$$

$$(e) \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx; \int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

Note 3: If the function f is **odd**, i.e, if $f(-x) = -f(x)$, for all x in $[-a, a]$, then

$$\int_{-a}^a f(x) dx = 0$$

Note 4: If the function f is **even**, i.e, if $f(-x) = f(x)$, for all x in $[-a, a]$, then

Summary

$$\int_{-a}^a f(x) \, dx = 2 \cdot \int_0^a f(x) \, dx$$

Part II

Recitation Questions

Problem 1

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Problem 1 Consider the following limit of Riemann sums of a function g on $[a, b]$:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n (x_k^* + \cos(x_k^*)) \Delta x, [0, \pi].$$

Express the limit as a definite integral. Use geometry to evaluate the resulting definite integral.

Explanation. We have

$$\int_0^\pi (x + \cos(x)) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n (x_k^* + \cos(x_k^*)) \Delta x.$$

To evaluate this definite integral, we'll start by writing $\int_0^\pi (x + \cos(x)) dx = \int_0^\pi x dx + \int_0^\pi \cos(x) dx$.

The first term is the definite integral $\int_0^\pi x dx$ which is the net area under the graph of $y = x$ on the interval $[0, \pi]$. This is a triangle of base π and height π , so it has area $\frac{1}{2}\pi^2$.

The second term $\int_0^\pi \cos(x) dx$ is the net area between the graph of $y = \cos(x)$ and the x -axis on the interval $[0, \pi]$. In the interval $[0, \frac{\pi}{2}]$ the function is positive, and in the interval $[\frac{\pi}{2}, \pi]$ the function is negative. By the symmetry of the cosine function, these two regions have the same geometric area, so the net area is zero.

$$\int_0^\pi (x + \cos(x)) dx = \int_0^\pi x dx + \int_0^\pi \cos(x) dx = \frac{\pi^2}{2}.$$

Problem 2

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Problem 2 Let $f(x)$ and $g(x)$ be functions for which we only know the following:

$$\int_1^4 f(x) dx = 7 \quad \int_2^4 f(x) dx = 5 \quad \int_1^4 g(x) dx = 2$$

Compute the following integrals, if possible. If it is not possible, give examples explaining why not.

(a) $\int_1^4 (8f(x) - 7g(x)) dx$

Explanation.

$$\begin{aligned} \int_1^4 (8f(x) - 7g(x)) dx &= 8 \int_1^4 f(x) dx - 7 \int_1^4 g(x) dx \\ &= 8(7) - 7(2) \\ &= 56 - 14 = 42 \end{aligned}$$

(b) $\int_1^2 (-f(x)) dx$

Explanation. First notice that

$$\int_1^4 f(x) dx = \int_1^2 f(x) dx + \int_2^4 f(x) dx$$

Therefore,

$$\int_1^4 f(x) dx - \int_2^4 f(x) dx = \int_1^2 f(x) dx$$

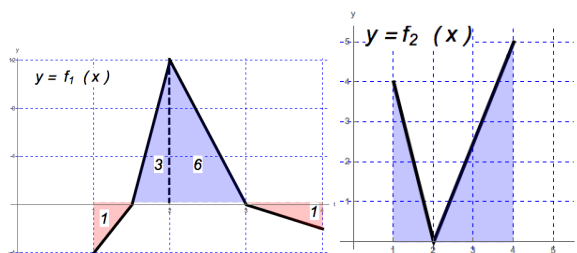
So

$$\begin{aligned} \int_1^2 (-f(x)) dx &= - \int_1^2 f(x) dx \\ &= - \left(\int_1^4 f(x) dx - \int_2^4 f(x) dx \right) \\ &= -(7 - 5) = -2. \end{aligned}$$

(c) $\int_1^4 |f(x)| dx$

Explanation. We are not given enough information to solve this integral because we do not know the regions where f is positive or negative. Consider the following two functions $f_1(x)$ and $f_2(x)$:

Problem 2



Just using geometry, one can check that

$$\int_1^4 f_1(x) dx = 7 \quad \int_2^4 f_1(x) dx = 5 \quad \int_1^4 f_2(x) dx = 7 \quad \int_2^4 f_2(x) dx = 5$$

and so both f_1 and f_2 satisfy the assumptions of f . But notice that

$$\int_1^4 |f_1(x)| dx = 11 \quad \text{and} \quad \int_1^4 |f_2(x)| dx = 7$$

These two examples demonstrate that we were not given enough information to solve this problem.

(d) $\int_1^4 (2 - x + f(x)) dx$

Explanation. First notice that since the integral is linear over addition:

$$\int_1^4 (2 - x + f(x)) dx = \int_1^4 2 dx - \int_1^4 x dx + \int_1^4 f(x) dx = \int_1^4 2 dx - \int_1^4 x dx + 7. \quad (1)$$

By using geometry, we can see that

$$\int_1^4 2 dx = 2(4 - 1) = 6$$

$$\int_1^4 x dx = 1(4 - 1) + \frac{1}{2}(4 - 1)(4 - 1) = 3 + \frac{9}{2} = \frac{15}{2}.$$

Then substituting into equation (1) gives:

$$\int_1^4 (2 - x + f(x)) dx = 6 - \frac{15}{2} + 7 = \frac{11}{2}.$$

Problem 3

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Problem 3 Evaluate the following sums:

$$(a) \sum_{k=1}^4 k^5$$

Explanation. $\sum_{k=1}^4 k^5 = 1^5 + 2^5 + 3^5 + 4^5 = 1 + 32 + 243 + 1024 = 1300.$

$$(b) \sum_{k=1}^{400} (5(k+1)^2 + 3)$$

Explanation.

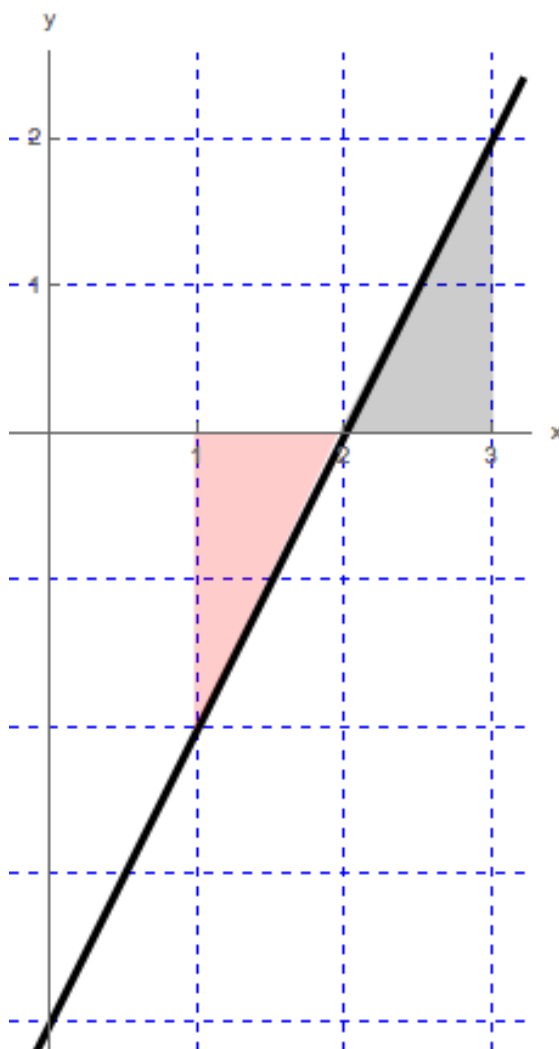
$$\begin{aligned} \sum_{k=1}^{400} (5(k+1)^2 + 3) &= \sum_{k=1}^{400} (5(k^2 + 2k + 1) + 3) \\ &= \sum_{k=1}^{400} (5k^2 + 10k + 8) \\ &= \sum_{k=1}^{400} 5k^2 + \sum_{k=1}^{400} 10k + \sum_{k=1}^{400} 8 \\ &= 5 \sum_{k=1}^{400} k^2 + 10 \sum_{k=1}^{400} k + 8(400) \\ &= 5 \left(\frac{400(400+1)(2(400)+1)}{6} \right) + 10 \left(\frac{400(400+1)}{2} \right) + 3,200 \\ &= 5(200)(401)(267) + 10(200)(401) + 3,200 \\ &= 107,067,000 + 802,000 + 3,200 = 107,872,200 \end{aligned}$$

Problem 4

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Problem 4 Use geometry to evaluate the definite integral. Sketch the graph of the function and shade the relevant regions.

(a) $\int_1^3 (2x - 4) dx$



Explanation.

We want to find the net area of the shaded region. We have two identical triangles, each with base 1 and height 2. Since the triangles will have identical area and one lies above and the other below the x -axis, the total area will be 0.

Therefore, $\int_1^3 (2x - 4) dx = 0$

(b) $\int_1^3 |2x - 4| dx$

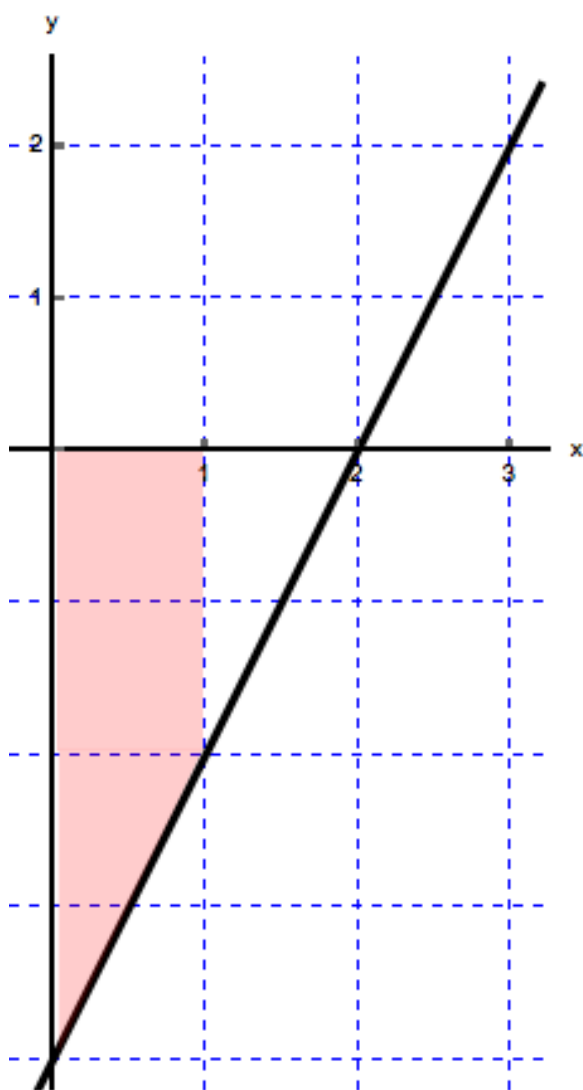
Explanation. The function changes the sign at $x = 2$. Therefore, we will split the integral into two integrals:

$$\int_1^3 |2x - 4| dx = \int_1^2 |2x - 4| dx + \int_2^3 |2x - 4| dx.$$

Since the function is negative for $x < 2$ and positive for $x > 2$, we write

$$\int_1^3 |2x - 4| dx = -\int_1^2 (2x - 4) dx + \int_2^3 (2x - 4) dx = 1 + 1 = 2.$$

(c) $\int_0^1 (2x - 4) dx$



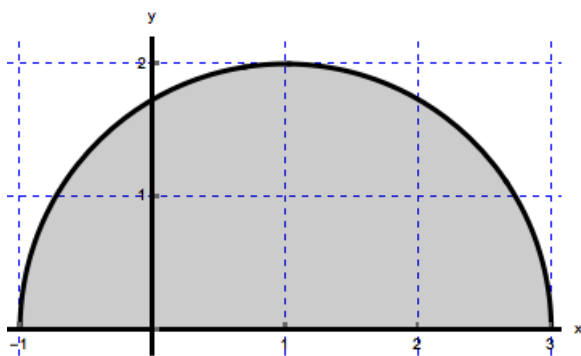
Explanation.

Here we need to find the area of a trapezoid. The trapezoid is made up of a rectangular top with area $(2)(1) = 2$ and triangle bottom with area $(1/2)(1)(2) = 1$. This gives us an area of 3, but, since f is negative on the interval $(0, 1)$, the value of the integral is the negative of this area.

Therefore, $\int_0^1 (2x - 4) dx = -3$

(d) $\int_{-1}^3 \sqrt{4 - (x - 1)^2} dx$

Problem 4



Explanation.

We are looking for the area of the entire semi-circle above the x -axis. This is a circle with radius 2 so we have $(1/2)\pi 2^2 = 2\pi$

$$\int_{-1}^3 \sqrt{4 - (x - 1)^2} dx = 2\pi$$

Problem 5

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Problem 5 (a) If f is an odd function, why is it true that $\int_{-a}^a f(x) dx = 0$?

Support your reasoning with a picture.

Explanation. If f is odd, then the regions between the graph of f and the x -axis from $[-a, 0]$ and $[0, a]$ are reflections of each other through the origin. Thus, these two regions will have the same area but with opposite signs since they are on opposite sides of the x -axis. They will therefore cancel each other out.

(b) If f is an even function, why is it true that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$?

Support your reasoning with a picture.

Explanation. If f is even, then the regions between the graph of f and the x -axis from $[-a, 0]$ and $[0, a]$ are reflections of each other through the y -axis. Thus, these two regions will have the same area with the same sign since they are on the same sides of the x -axis. So you can only find one of these areas and then double it.

Problem 6

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Problem 6 (a) Find the following definite integral:

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx$$

Explanation. Let $f(x) = \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}}$. Then notice that

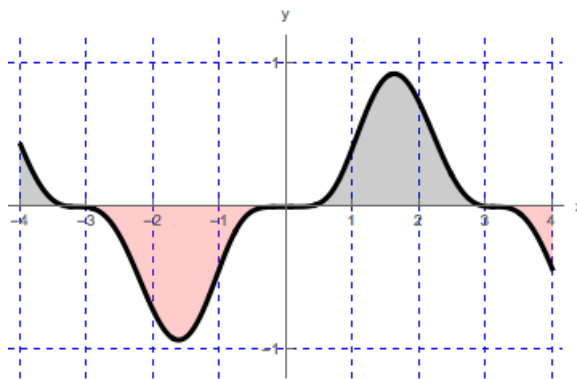
$$\begin{aligned} f(-x) &= \frac{(-x)^2 \sin^3(-x)}{\sqrt{(-x)^4 + 1}} \\ &= \frac{x^2 (-\sin(x))^3}{\sqrt{x^4 + 1}} \quad (\text{since } \sin(x) \text{ is an odd function}) \\ &= \frac{-x^2 \sin^3(x)}{\sqrt{x^4 + 1}} \\ &= -f(x). \end{aligned}$$

Thus, f is an odd function and therefore

$$\int_{-4}^4 \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}} dx = 0.$$

We can illustrate our computation with the graph of the function f , where

$$f(x) = \frac{x^2 \sin^3(x)}{\sqrt{x^4 + 1}}, \text{ for } -4 \leq x \leq 4.$$



(b) Suppose that f is an even function. Given that $\int_0^6 f(x) dx = 13$, find

$$\int_{-6}^6 (5f(x) + 14) dx.$$

Explanation. First notice that

$$\int_{-6}^6 (5f(x) + 14) dx = 5 \int_{-6}^6 f(x) dx + \int_{-6}^6 14 dx. \quad (2)$$

Problem 6

Since f is even, we know that

$$\int_{-6}^6 f(x) dx = 2 \int_0^6 f(x) dx = 2(13) = 26. \quad (3)$$

We also know that $g(x) = 14$ is also an even function. We have: $\int_{-6}^6 14 dx =$

$$2 \int_0^6 14 dx$$
$$2 \int_0^6 14 dx = 2(14)(6) = 168. \quad (4)$$

Then substituting equations (3) and (4) into equation (2) gives:

$$\int_{-6}^6 (5f(x) + 14) dx = 5(26) + 168 = 130 + 168 = 298.$$

Problem 7

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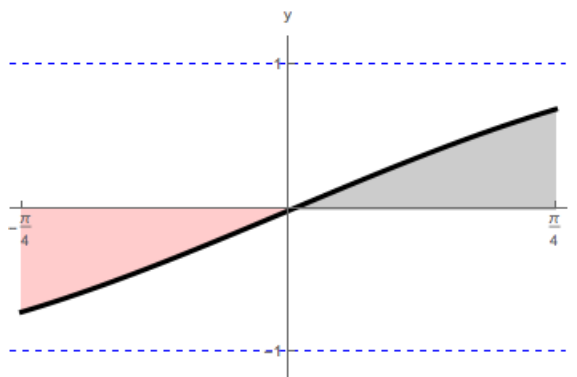
Problem 7 Evaluate the following integrals using symmetry arguments.

(a) $\int_{-\pi/4}^{\pi/4} \sin(t) dt$

Explanation. Since the function $f(t) = \sin(t)$ is odd, $\int_{-\pi/4}^{\pi/4} \sin(t) dt = 0$

We can illustrate our computation with the graph of the function f , where

$$f(t) = \sin t, \text{ for } -\frac{\pi}{4} \leq t \leq \frac{\pi}{4}.$$



(b) $\int_{-2}^2 (1 + x + 3x^7 - x^9) dx$

Explanation. In order to take advantage of symmetry, we will split the given polynomial into a sum of its even and odd parts. We will then integrate each part separately:

$$\int_{-2}^2 (1 + x + 3x^7 - x^9) dx = \int_{-2}^2 1 dx + \int_{-2}^2 (x + 3x^7 - x^9) dx$$

We will first integrate an even function:

$$\int_{-2}^2 1 dx = 2 \int_0^2 1 dx = 2(2) = 4$$

Then an odd function:

$$\int_{-2}^2 (x + 3x^7 - x^9) dx = 0$$

Therefore: $\int_{-2}^2 (1 + x + 3x^2 - x^9) dx = 4$

(c) $\int_{-\pi}^{\pi} x \cos(x) dx$

Problem 7

Explanation. $\int_{-\pi}^{\pi} x \cos(x) dx = 0$, since the product of an odd and even function is odd.

We can illustrate our computation with the graph of the function f , where

$f(x) = x \cos x$, for $-\pi \leq x \leq \pi$.

